

MODELLING UPDATE · WORK IN PROGRESS

Modelling Enhanced Rock Weathering in Sub-Saharan Africa

An evolving, spatially explicit model of where spreading crushed basalt on acid cropland might pay farmers, the climate, or **both together**. Everything shown here is a draft: we keep revising the model as better data and evidence become available, and several elements still need research in geochemistry, yield response, and markets and economics.

IN DIALOGUE WITH

CASCADE CLIMATE Geochemistry and yield-benefit modelling; comparison with their cost and MRV tools.	YALE UNIVERSITY Weathering geochemistry and carbon accounting; assumptions checked against the Yale-led 2026 review.	MATI CARBON Yield-benefit modelling, drawing on their smallholder deployments.
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TEAM

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FUNDING TO DATE

Gates Foundation, through CIAT

STATUS

Working draft; model and paper under active revision

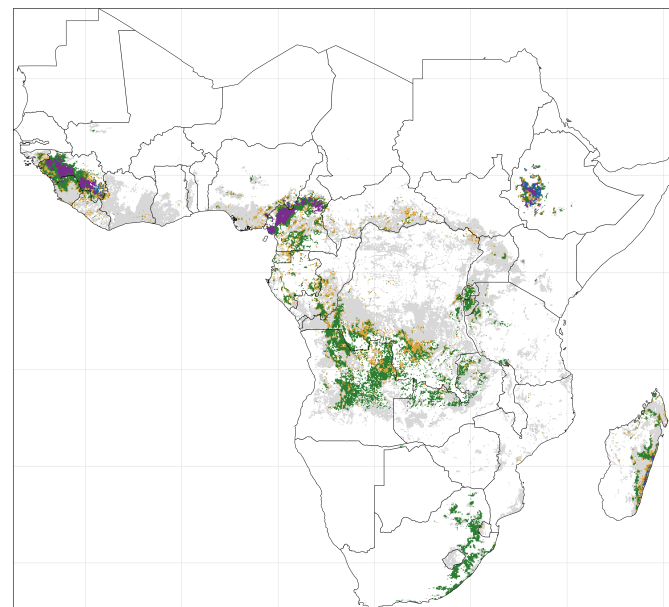
ASK OF THIS MEETING

Views on priorities for grounding the work

WHAT THE MODEL DOES, AND WHAT IT SUGGESTS SO FAR

Early indications: the two benefits rarely pay alone, but often pay together.

Return typology · equilibrium, net-export CDR, targeted



both private only public only combined only neither

Crushed basalt acts like agricultural lime, raising yields on acid soils, and it also removes carbon dioxide that could be sold as a credit. For every patch of cropland the model asks which of these benefits would cover the full cost of the rock, using public soil, crop, climate, geology, road and price data.

PRELIMINARY PATTERN

In the current draft, land where yield alone pays is larger than land where carbon alone pays, and a substantial area appears viable only when the two are combined. If this holds up, carbon finance would be acting as a cost-share that unlocks a yield investment smallholders cannot otherwise afford, with the yield gain staying on the farm.

How to read it. The pattern is more trustworthy than any number on the map. The absolute areas move every time we update an input, and several inputs are still uncertain (next slides).

Green: yield pays alone · blue: carbon pays alone · purple: both · gold: only together · grey: neither. Draft, subject to revision.

HOW THE WORK HAS EVOLVED SINCE MAY 2026

An iterative build: each round replaces an assumption with better evidence.

MAY · FIRST VERSION

A working map

Public data layers (soils, crops, rock geology, roads, climate, prices) brought into one model covering 44 countries and 23 crops. Many placeholders and first-guess constants.

JULY · CARBON ACCOUNTING

More conservative carbon

We stopped counting carbon that is used up neutralizing soil acidity, a step most global models skip. This reduced early-year carbon sharply and changed how the carbon case should be read.

SEPTEMBER · EVIDENCE CHECK

Assumptions re-examined

Key assumptions compared against the 2026 systematic review of the field and revised where the evidence pointed elsewhere (temperature response, soil-pH response, yield bounds). Model re-run; numbers moved.

NOW

Ongoing revision

Draft paper in preparation. Sensitivity runs show which inputs matter most, which sets the research agenda. We expect further rounds as trial data and partner input arrive.

WHAT WE ARE TRYING TO DO

Use the best available data at each step and be explicit about what is assumed. Every constant in the model is catalogued with its source, and every result carries a range rather than a point.

WHAT THAT MEANS FOR THE NUMBERS

They have moved with every revision and will move again. The qualitative ordering (yield case larger than carbon case; a meaningful combined-only area) has been stable so far, but we present it as a hypothesis to test, not a finding.

WHAT STILL NEEDS RESEARCH AND EXPLORATION

Open questions in three areas.

GEOCHEMISTRY

How much carbon a tonne of rock really removes

- Removal rates in tropical smallholder soils are largely unmeasured; ours are scaled from temperate evidence.
- How much carbon is lost or delayed in the soil, in rivers and at the ocean, and over what timescales.
- How much re-precipitates in drier soils; today a rule of thumb by aridity band.
- Which rock chemistry and grind size to assume, given weak reporting in the literature.

YIELD RESPONSE

How much harvest farmers actually gain

- The evidence base is roughly a dozen trials, few of them in Africa and few longer than a season or two.
- The crop response is currently a suitability curve; a trial-anchored statistical model is built but not yet run.
- Whether too much rock can depress yields, and by how much, is not quantified anywhere.
- Responses in the new Ghana and Kenya trials still need to be folded in.

MARKETS AND ECONOMICS

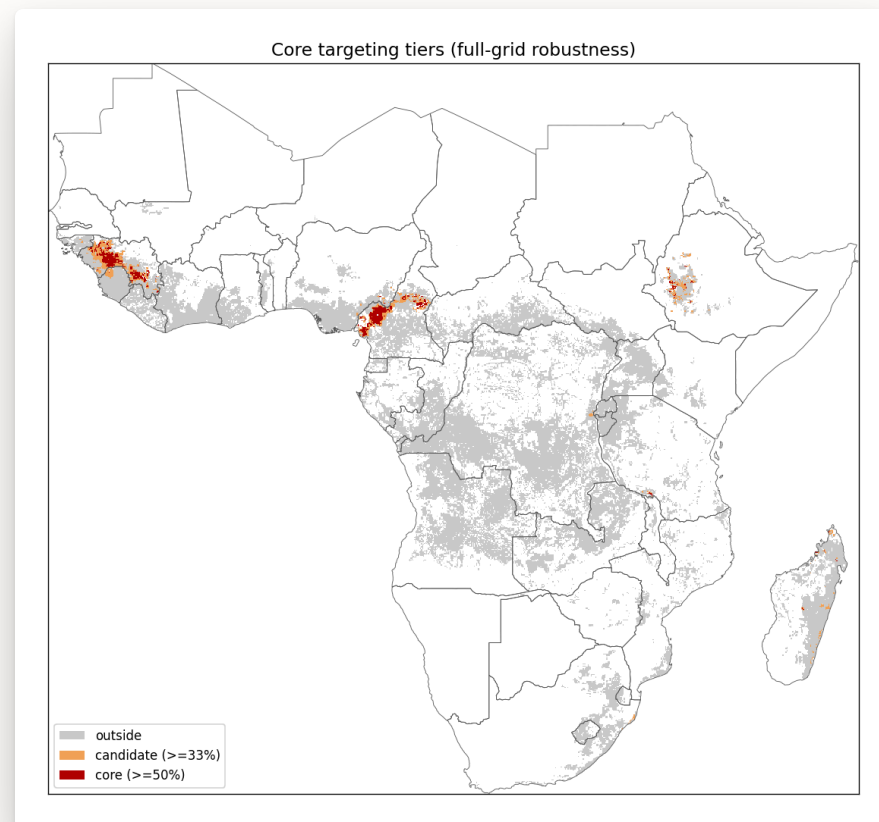
Whether and how it could be financed

- Carbon price and monitoring cost are assumptions; the credit market is small and buyers and suppliers disagree on price.
- Delivered cost is modelled from a quarry price and trucking rate, not from quotes on the ground.
- Who finances the first dose (tenure, credit, cost-sharing) is not represented at all.
- How registries would treat additionality and baselines for smallholder deployments.

Sensitivity runs so far suggest the carbon-removal rate per tonne is the input the results are most sensitive to, and the yield response the least; both remain open.

WHERE WE ARE HEADING, AND WHAT WOULD MAKE THE MODELLING MORE GROUNDED

Keep improving the model, and start testing it in the field.

**NEXT ON THE MODELLING SIDE**

Fold in new trial data and partner input; run the trial-anchored yield model; align the carbon accounting with what credit registries require; keep the model, code and maps open so others can reproduce and challenge them; and complete the draft paper.

GROUND TRUTH THAT WOULD HELP MOST

Multi-year field measurements of carbon removal in tropical smallholder soils; real quarry, grinding and trucking costs near candidate rock sources; and farm-level evidence on yield, adoption and who can pay. The map (red and orange, mainly the Cameroon and Guinea highlands) suggests where such measurements would be most informative, though that suggestion is itself a draft.

PARTNERSHIPS THAT WOULD HELP

National soil and agricultural research institutes in the candidate countries; a credit registry to co-design accounting; project developers and buyers willing to share deployment and monitoring data; and rock suppliers near the candidate sources.